GATE ALL BRANCHES



General Aptitude

SPATIAL APTITUDE

Lecture No.- 01



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Recap of Previous Lecture



Topic Venn Diagrams & Dice

Topics to be Covered



Topic-1 Formation of Images

[MCQ]

#Q. The six faces of a dice have been marked with alphabets A, B, C, D, E and F respectively. This dice is rolled down three times. The three positions are shown as: Find the alphabet opposite A.



A C

B D

C E

D F

E ⇒ C, D, B, F

E ↔ A

(B)

(C)

Assignment

[MCQ]

#Q. Two positions of a dice are shown below. If the face with 1 dot is at the bottom, then the number of dots on the top is

A 2

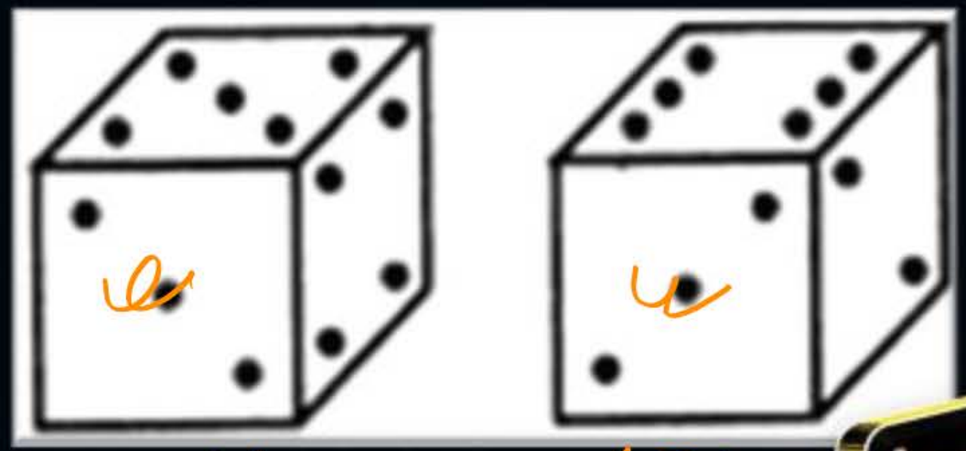
B 3

C 4

D 5

3 ⇒ 5, 4, 2, 6

3 ⇔ 1



Assignment

[MCQ]

#Q. A cube has six different symbols drawn over its six faces. The symbols are dot, circle, triangle, square, cross and arrow. Three different positions of the cube are shown in figures X, Y, and Z.

Which symbol is opposite the dot?



(x) (y) (z)

A Circle

B Arrow

C Cross

D Triangle

X ⇒ Square, triangle

Circle, Arrow

Cross ↔ dot

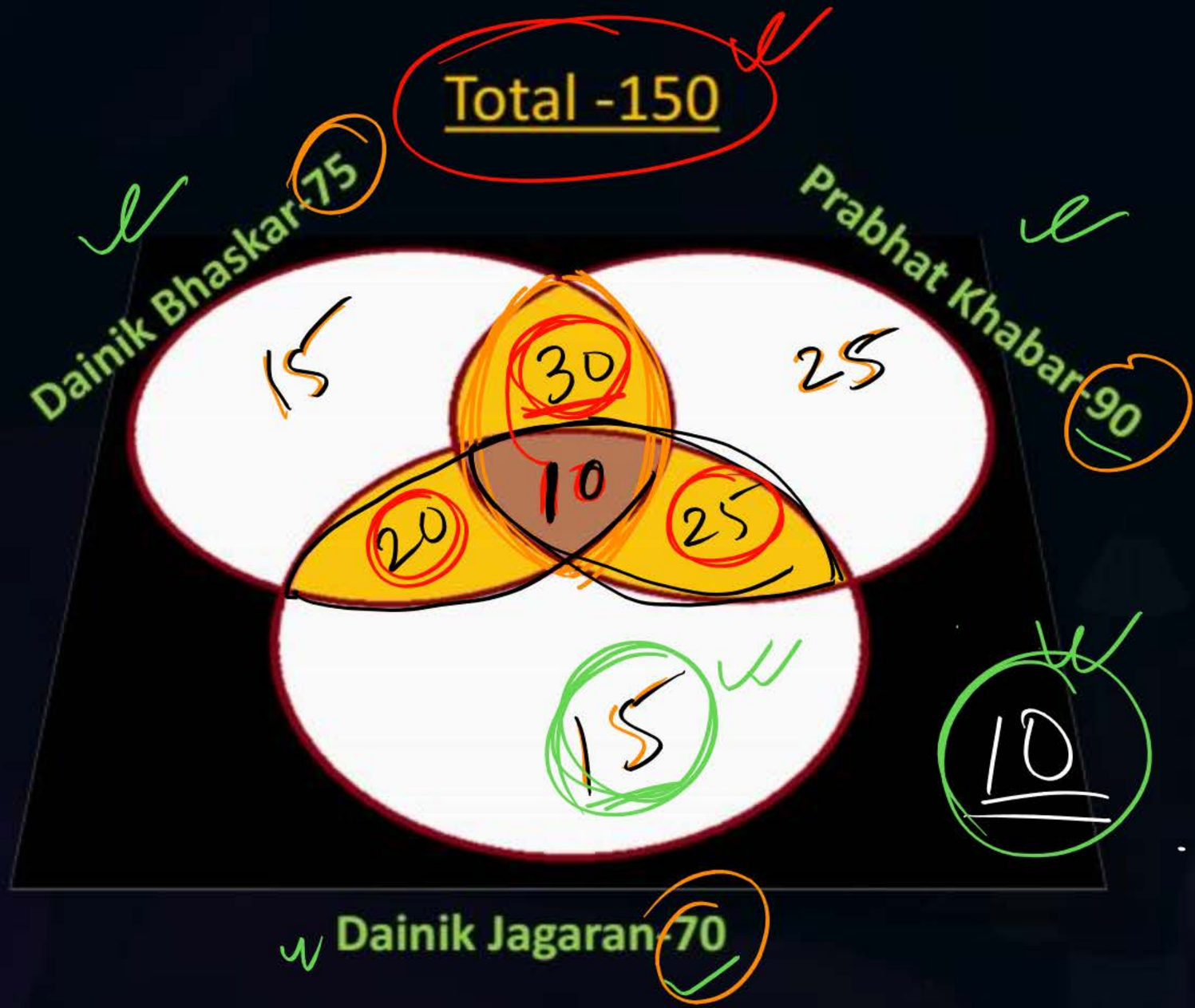
[MCQ]

#Q. In a survey of 150 readers it has been found that 75 read Dainik Bhaskar, 90 read Prabhat Khabar and 70 read Dainik Jagaran. 40 read Dainik Bhaskar and Prabhat Khabar; 35 read Prabhat Khabar and Dainik Jagaran; 30 read Dainik Bhaskar and Dainik Jagaran and 10 read all the three.



(i) How many read exactly two newspapers?

(ii) How many read neither Dainik Bhaskar nor Prabhat Khabar?


Total -150
Dainik Bhaskar-75
Prabhat Khabar-90
Dainik Jagaran-70
15
30
25
10
20
25
15
10

[MCQ]

#Q. In a class with 1000 students, 600 play chess, 300 play hockey and 50 play both the games. The number of students neither play chess nor hockey is

____

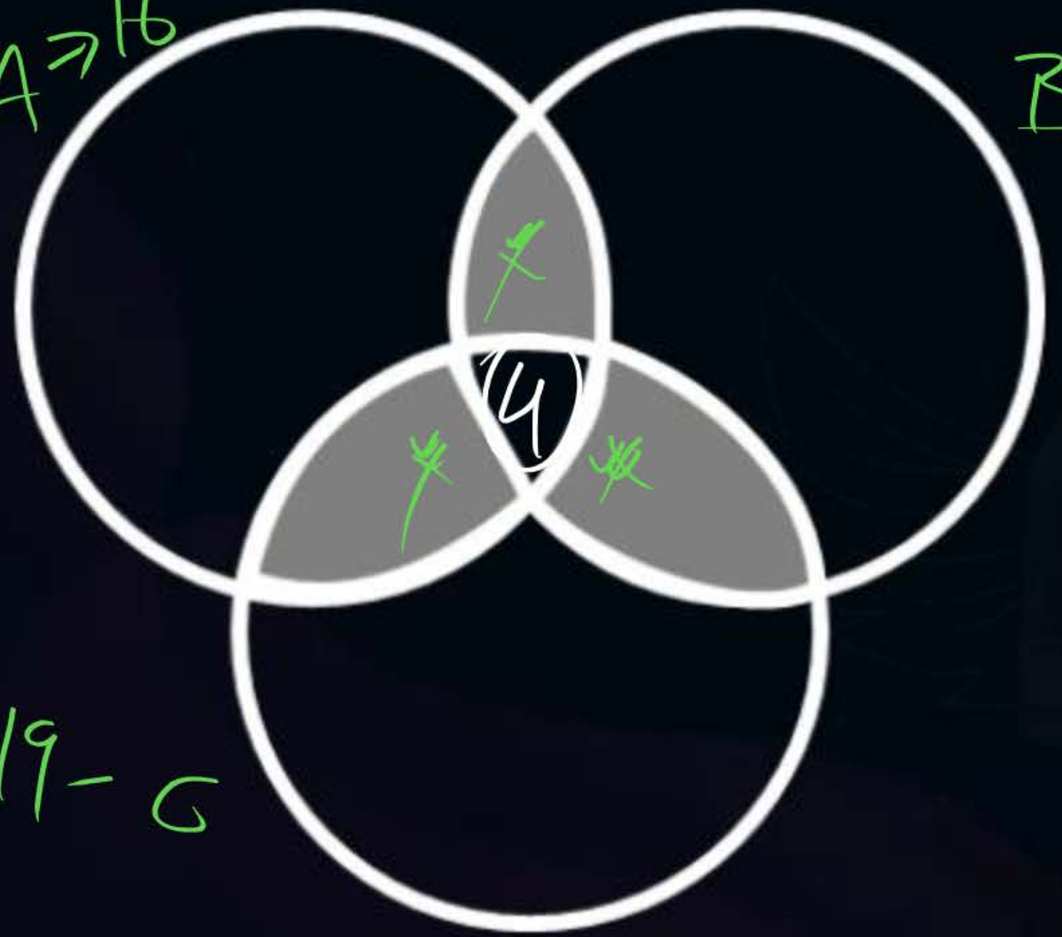


Assignment

[MCQ]

48 − 40 = 8 $\frac{8}{2} = 4$

#Q. There are 40 students in a class. They have to watch movies from A, B and C. They either watch one or all three movies. 16 students watch A, 13 students watch B, 19 students watch C. How many students watched all three movies?



Assignment

16
+ 13
+ 19
48

4

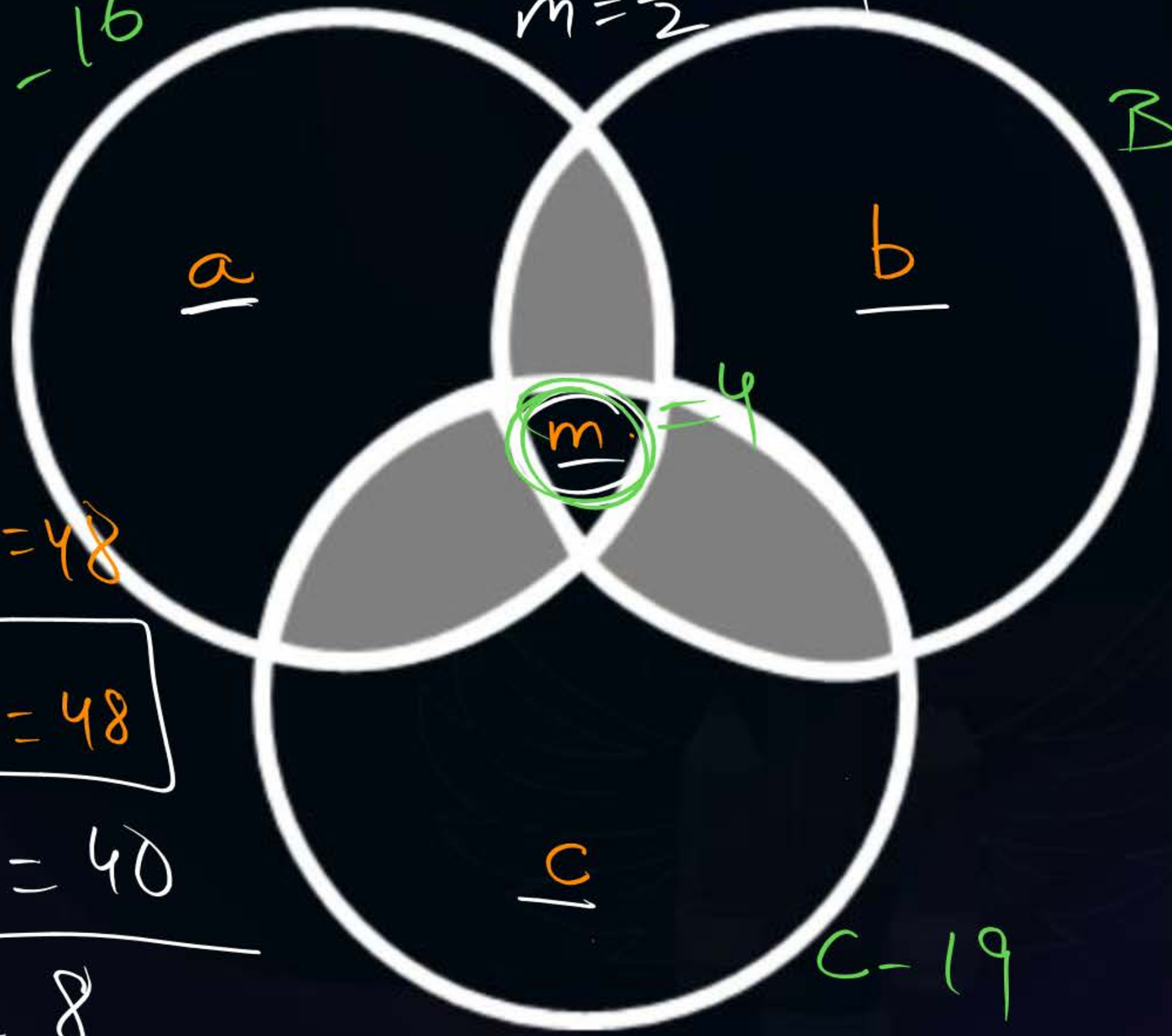
40 Ans

A - 16

$m = \frac{8}{2} = 4$

B - 13

C - 19



$16 + 13 + 19 = 48$

$a + m + b + m + c + m = 48$

$$a + b + c + 3m = 48$$

$$a + b + c + m = 40$$

$$2m = 8$$

[MCQ]

160 − 15 = 145

#Q. In a class of 160 students, it was found that 65 play Cricket 70 play Hockey and 90 play Football, 30 play Cricket and Hockey, 40 Cricket and Football, 35 play Hockey and Football and 15 play none of these three games.

↷ How many play all three games? 25

↷ How many students play exactly one game? 90



225
145
80

65
70
90
225

105
−80
25

[MCQ]

#Q. Given below are two statements followed by two conclusions. Which one of the following options are logically inferred?

Statement 1: Some actors are dancers.

Statement 2: All dancers are fat.

Conclusion I: Some actors are fat.

Conclusion II: All actors are fat.

A. Only I follow

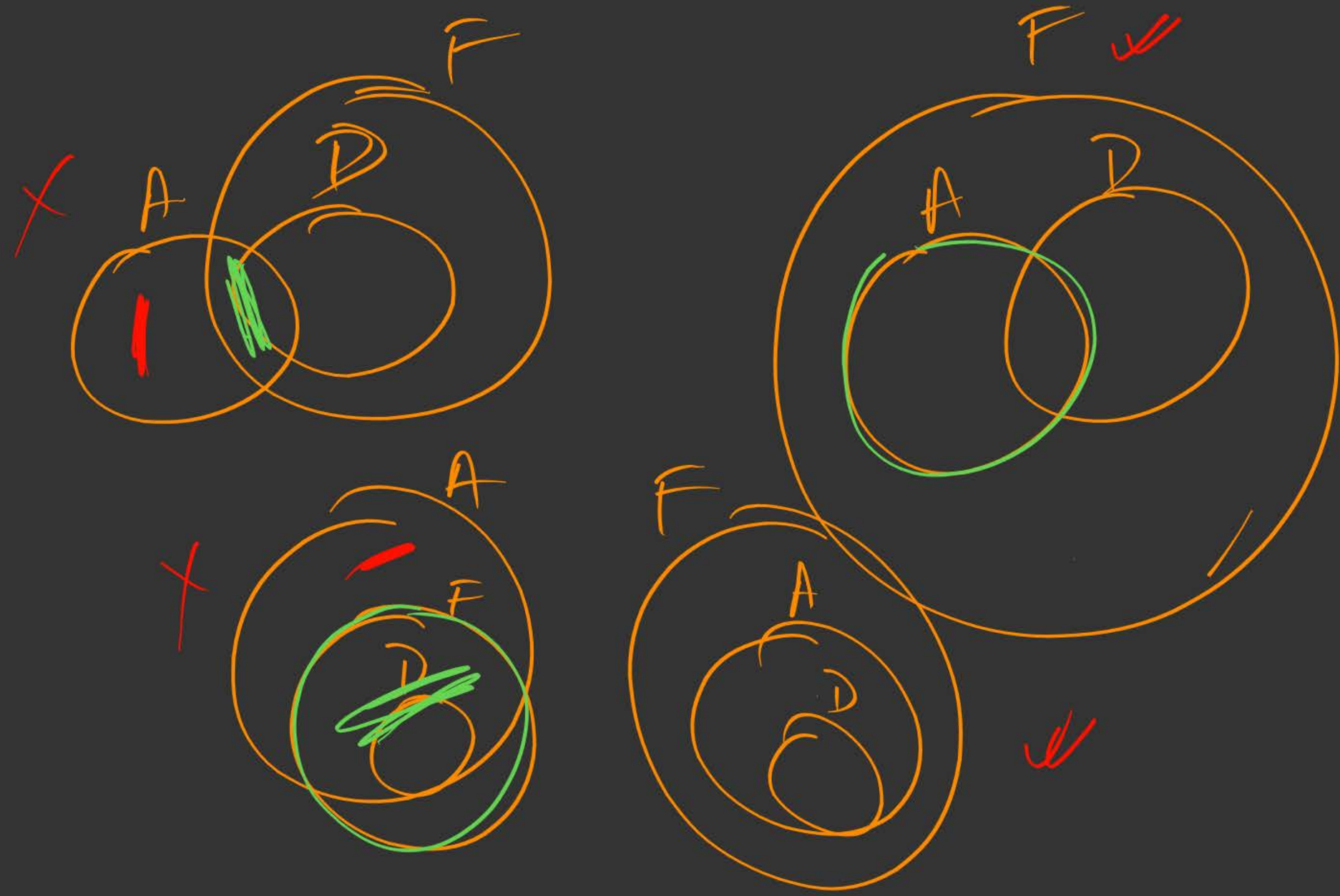
B. Only II follow

C. Both I and II follow

D. Neither I nor II follow

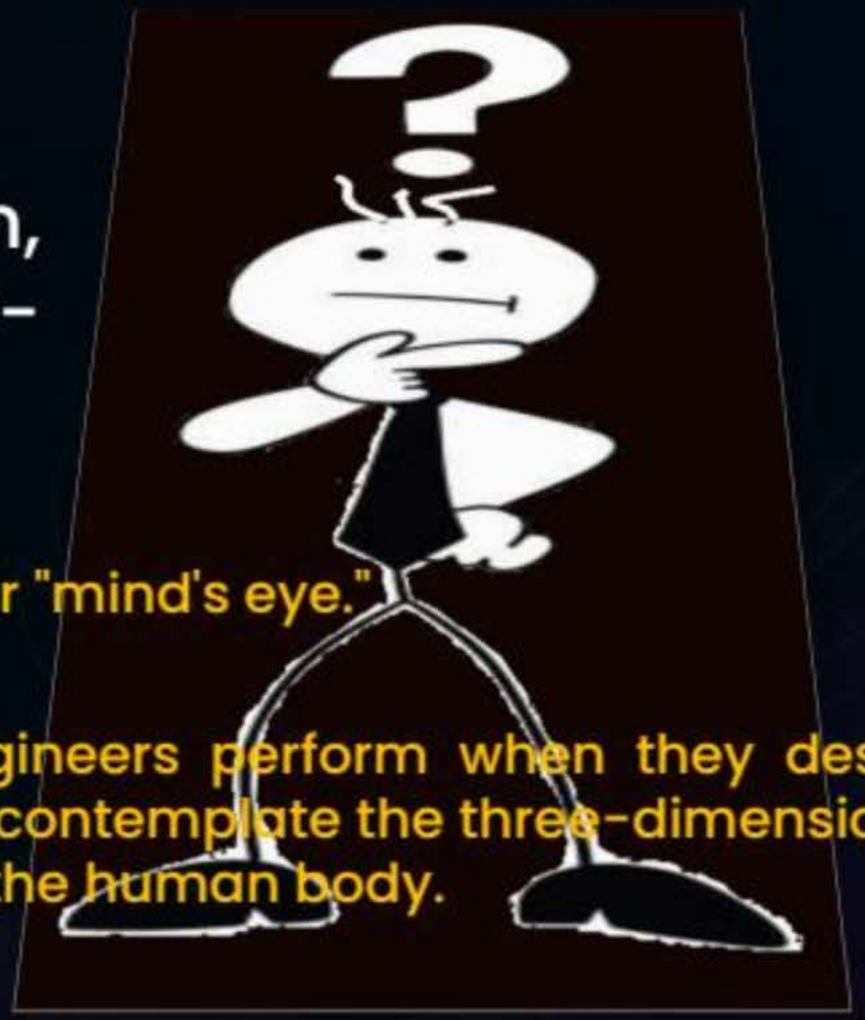





F
D
X
A
1
A
X
F
D
F
A
D
F
A
D

SPATIAL APTITUDE

"the ability to generate, retain, retrieve, and transform well-structured visual images"



- It's what we do when we visualize shapes in our "mind's eye."

- It's the mental feat that architects and engineers perform when they design buildings. The capacity that permits a chemist to contemplate the three-dimensional structure of a molecule, or a surgeon to navigate the human body.

Formation of Images

- In each of the following questions, group the given figures into three classes using each figure only once.

[MCQ]

#Q. Group the given figures into three classes using each figure only once.



A. 1,3,9 / 2,5,8 / 4,6,7

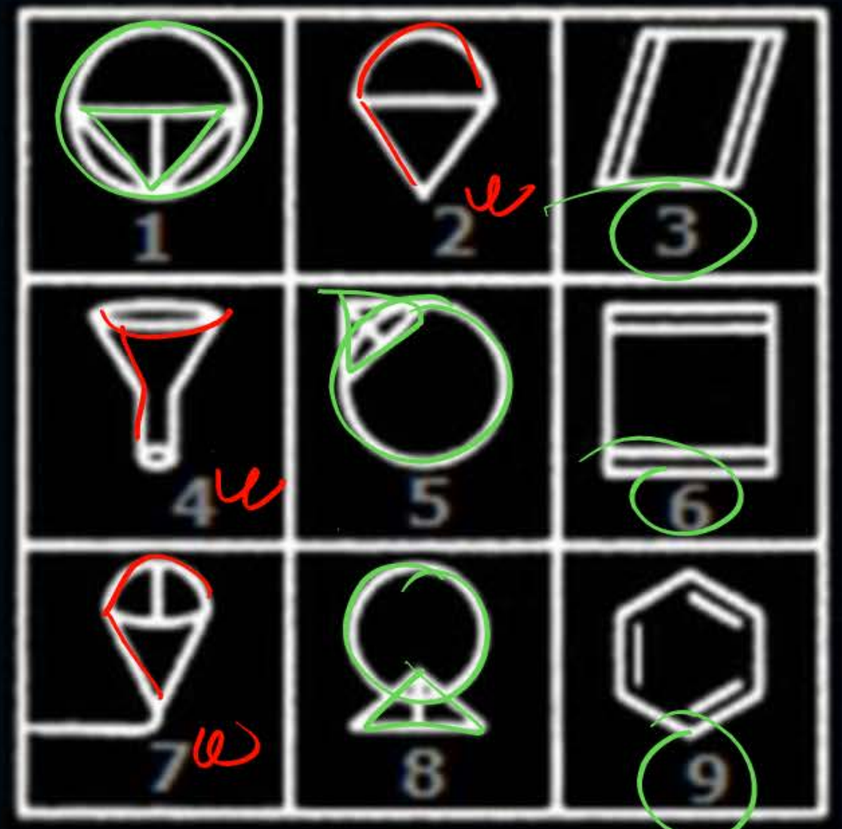
B. 1,5,8 / 4,6,7 / 2,3,9

C. 3,5,8 / 1,2,4 / 6,7,9

D. 2,5,9 / 1,3,8 / 4,6,7

[MCQ]

#Q. Group the given figures into three classes using each figure only once.



A 1,5,8 / 2,6,7 / 3,4,9

B 1,5,8 / 2,4,7 / 3,6,9

C 1,5,7 / 2,6,8 / 3,4,9

D 1,5,8 / 2,6,9 / 3,4,7

[MCQ]

#Q. Group the given figures into three classes using each figure only once.



A 1,4,8 / 2,7,9 / 3,5,6

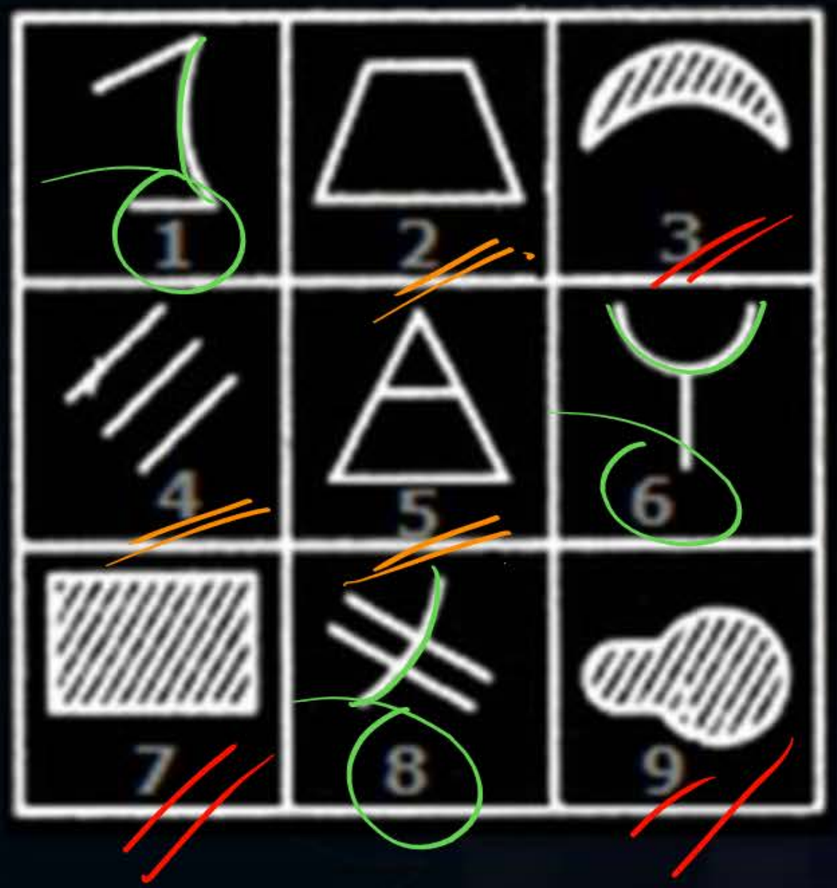
B 1,4,8 / 2,5,6 / 3,7,9

C 1,4,6 / 2,5,7 / 3,8,9

D 1,3,7 / 2,4,6 / 5,8,9

[MCQ]

#Q. Group the given figures into three classes using each figure only once.



A 1,3,6 / 4,5,8 / 2,7,9

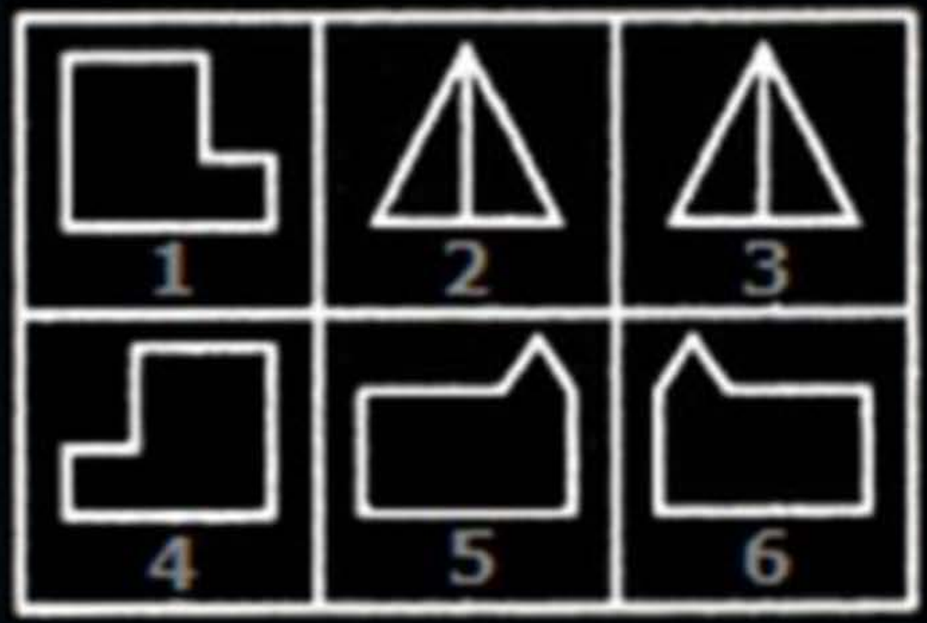
B 2,3,9 / 4,5,8 / 1,6,7

C 3,8,9 / 1,2,7 / 4,5,6

D 1,6,8 / 3,7,9 / 2,4,5

[MCQ]

#Q. Group the given figures into three classes using each figure only once.



A 1,4 / 2,3 / 5,6

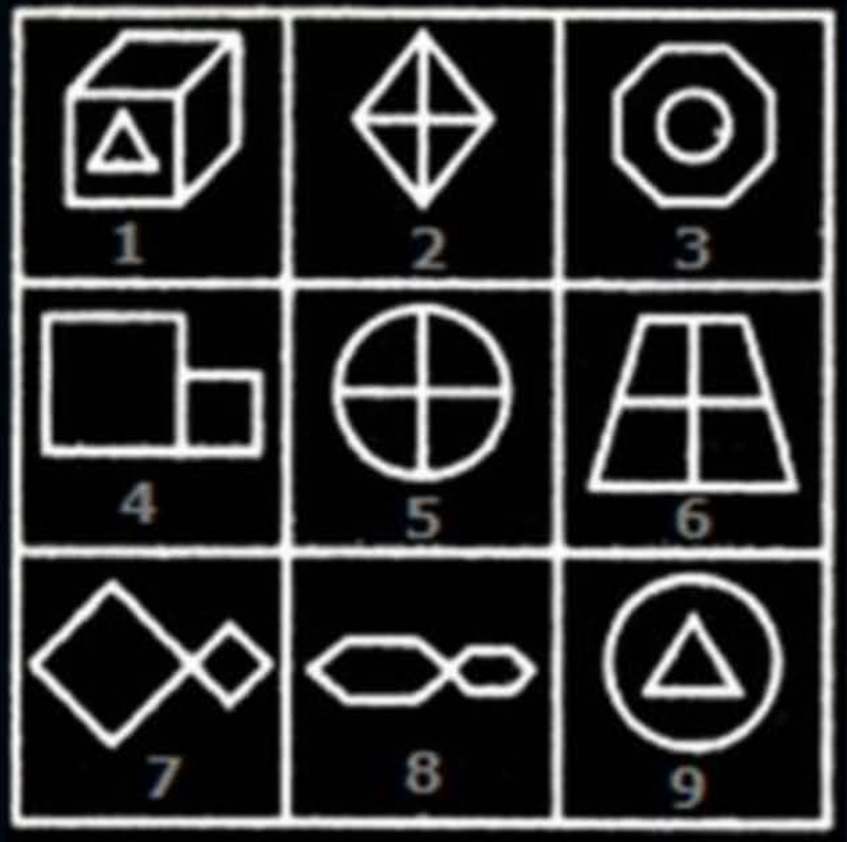
B 1,5 / 2,6 / 4,3

C 1,6 / 2,3 / 4,5

D 1,2 / 3,6 / 4,5

[MCQ]

#Q. Group the given figures into three classes using each figure only once.



Assignment

A 1,2,4 / 3,5,7 / 6,8,9

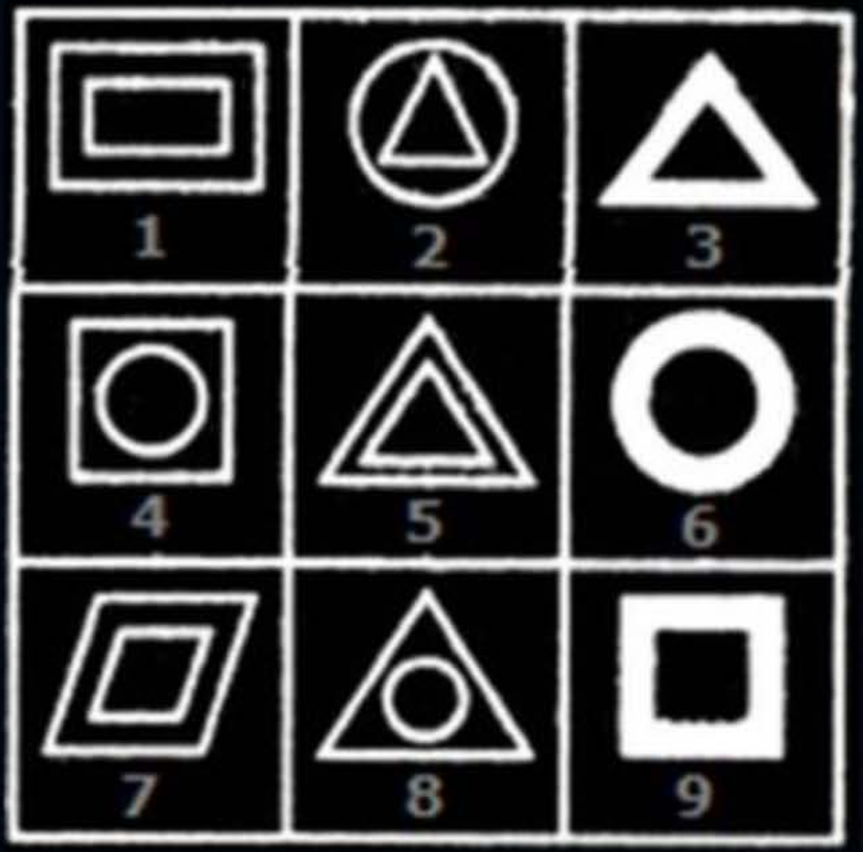
B 1,3,9 / 2,7,8 / 4,5,6

C 1,3,6 / 2,4,8 / 5,7,9

D 1,3,9 / 2,5,6 / 4,7,8

[MCQ]

#Q. Group the given figures into three classes using each figure only once.



Assignment

A 1,4,7 / 2,5,8 / 3,6,9

B 1,5,7 / 2,4,6 / 3,9,8

C 1,5,7 / 2,4,8 / 3,6,9

D 1,7,9 / 3,5,8 / 2,4,6

[MCQ]

#Q. Group the given figures into three classes using each figure only once.

Assignment



A 1, 3, 5 | 2, 4, 7 | 6, 8, 9

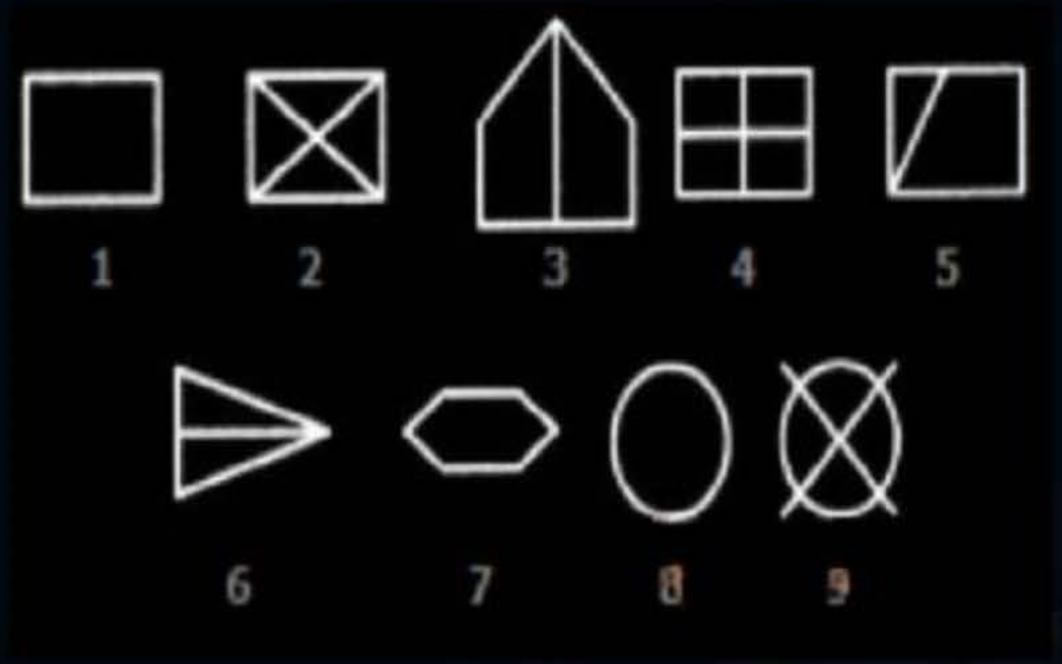
B 1, 5, 7 | 2, 3, 6 | 4, 8, 9

C 1, 6, 9 | 2, 4, 7 | 3, 5, 8

D 1, 3, 5 | 2, 6, 7 | 4, 8, 9

[MCQ]

#Q. Group the given figures into three classes using each figure only once.



Assignment

A. 1, 2, 4 | 3, 5, 6 | 7, 8, 9

B. 1, 3, 4 | 2, 8, 9 | 5, 6, 7

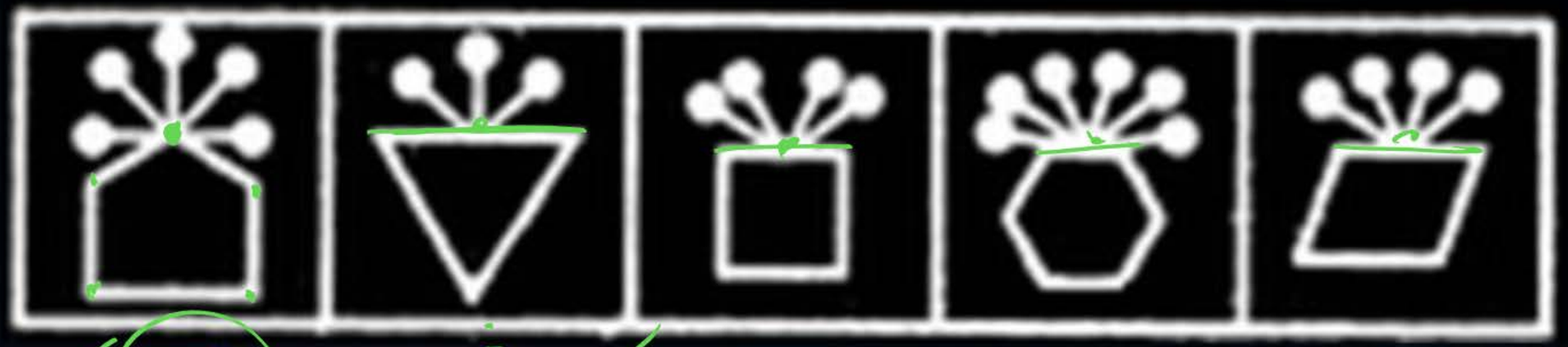
C. 1, 7, 8 | 3, 5, 6 | 2, 4, 9

D. 1, 7, 8 | 2, 3, 6 | 4, 5, 9

- In each problem, out of the five figures marked (1), (2), (3), (4) and (5), four are similar in a certain manner. However, one figure is not like the other four. Choose the figure which is different from the rest.

[MCQ]

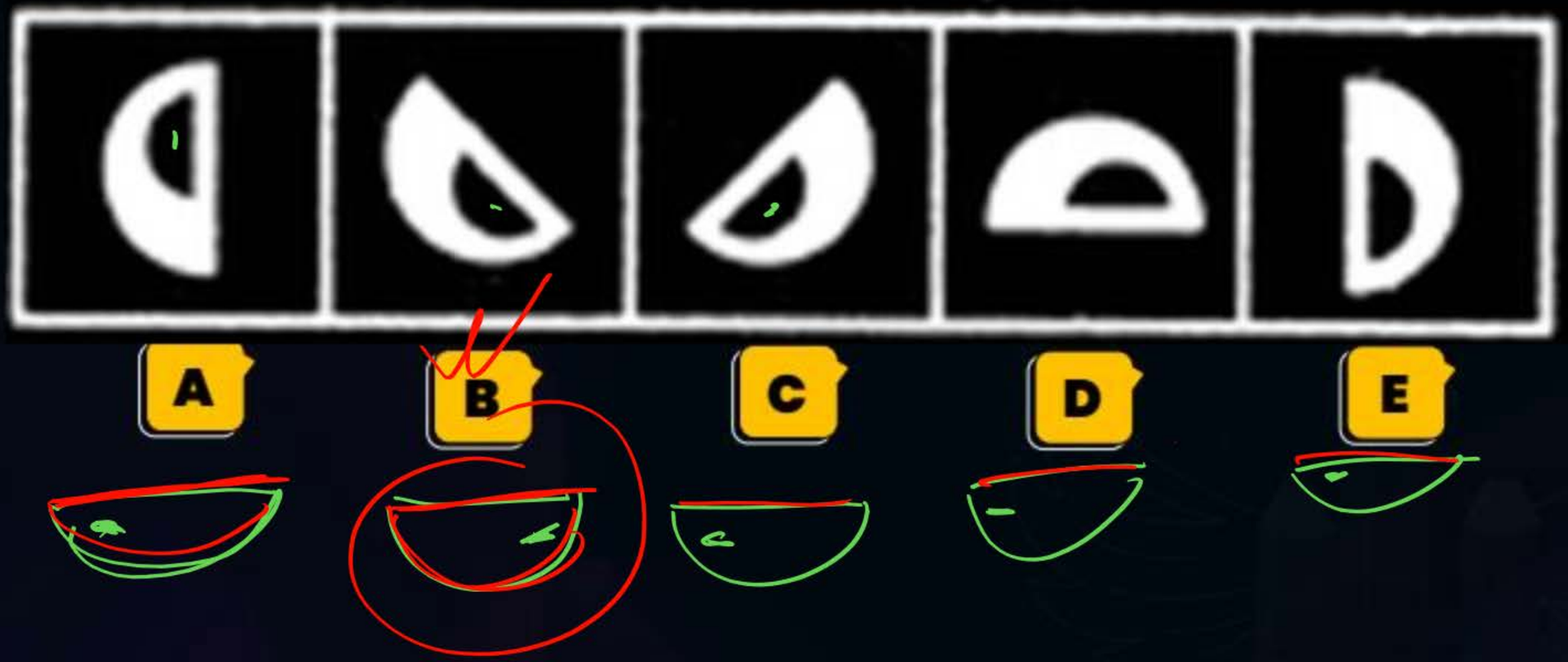
#Q. Choose the figure which is different from the rest.



A B C D E

[MCQ]

#Q. Choose the figure which is different from the rest.



[MCQ]

#Q. Choose the figure which is different from the rest.



A B C D E

[MCQ]

#Q. Choose the figure which is different from the rest.



A B C D E

Assignment

[MCQ]

#Q. Choose the figure which is different from the rest.



A B C D E

Assignment

[MCQ]

#Q. Choose the figure which is different from the rest.



A B C D E

Assignment

[MCQ]

#Q. Choose the figure which is different from the rest.



A B C D E

Assignment

[MCQ]

#Q. Choose the figure which is different from the rest.



A B C D E

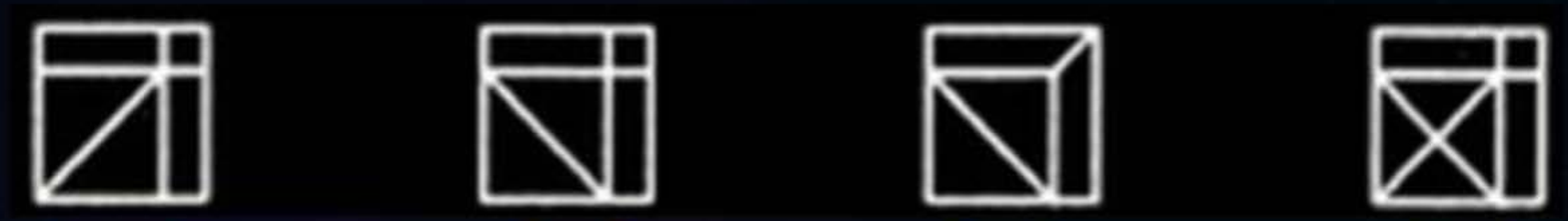
Assignment

[MCQ]

#Q. Identify the figure that completes the pattern.

Assignment





A B C D

[MCQ]

#Q. Identify the figure that completes the pattern.

Assignment





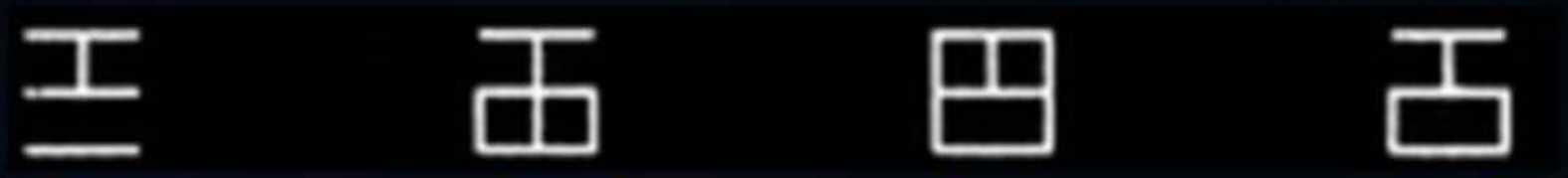
A B C D

[MCQ]

#Q. Identify the figure that completes the pattern.

Assignment





A B C D

[MCQ]

#Q. Identify the figure that completes the pattern.

Assignment





A B C D



2 mins Summary

Topic

Formation of Images

THANK - YOU